PT Daily Digest - 本日の注目トピック 5選（日本語全文レポート）

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#1 【学術論文】 The design of soft-actuated upper-body exoskeletons for movement assistance and physical therapy

著者: Daniel George	掲載誌: CentAUR (University of Reading) (2029-01-01)
DOI: 10.48683/1926.00129237	区分: オープンアクセス（全文確認可）

1. 論文・ニュース概要

None

2. 公開ステータス

- オープンアクセス区分: オープンアクセス（全文閲覧可能）
- 掲載誌: CentAUR (University of Reading)
- 出版日: 2029-01-01

3. 英文アブストラクト（原文抄録）

This thesis presents the design, implementation, andvalidationof novel soft-actuatedexoskeletons formovementassistanceandrehabilita tion.Motivatedbytheneedtoenhancequalityoflifeforindividualswith neurologicalimpairments,suchasstrokesurvivorsandspinal-cord-injured individuals, thisworkexploreshybridassistivedevices thatmerge the comfort,compliance,andweightofsoftactuatorswiththestability,relia bility,andalignmentprovidedbyrigidstructures.Twokeysystemswere introduced: (1)ahighdegree-of-freedom(DOF) soft-actuatedarmand spineexoskeletonthatuseslightweightpolyethylenepneumaticactuators integrated intoa3D-printed frame, achievingover sevenDOFs includ ingforearmandshouldersupination/pronationmovements,easilycarried andmobile, capableof 3-dimensionalmovement. Experimental evalua tionsdemonstrate that theorigami-inspiredsystemdelivers torquesup to4.46Nm, supports full 3Darticulation, andeffectivelyredistributes loadthroughasegmentedelasticspine, enablingwearablegravitycom pensationevenforuserswithlimitedstrength. (2)elastic-tensegrityjoints inspiredbybiologicaltensegritytounlockmulti-axisflexibilitywhilepre servingmechanical reliabilityanduser comfort. Byunlocking the full rangeof soft-actuatormotionandeliminating the rigidityand friction of traditional joints, thesedesigns substantiallyenhance comfort, flex ibility, andpassivestabilitythatdemonstrate intelligentphysicaldesign canoutperformcontrol-basedcompensation.Theelastic-tensegrityjoints, whencombinedwithelectromyography(EMG)andelectroencephalogra phy(EEG) control, achieve smooth, adaptivemotion, passivedamping of tremors, andaccurate intentdetection(EMGaccuracy100%; EEG upto82%, intentpredictionis improvedbyzero-crossingmethods),val idatingthe feasibilityofbi-manual rehabilitationandbrain-machine in terfacingwithout theelectromagnetic interferenceassociatedwithrigid actuators.Beyondhardwareinnovations,supportingstudiesincludemul timodal feedbackexperimentsthatenhancethesenseofownership(SoO) overvirtuallimbs,suggestingsynergisticpotentialforimmersivetherapies andaligningwiththebroaderthemeofreconnectingthebrain-bodyloop. Together,thesecontributionsaddressimportantgapsincurrentassistive technologiesandthehuman-machineinterface(HMI);affordability,acces sibility,comfort,andadaptability,aswellasopeningupanewwholenew generationofexoskeletondesignsforclinicalandat-homerehabilitation.

4. 原文確認のご案内

研究の詳細データや査読全文については、必ず上記原文URLよりご確認ください。

本レポートはAI（Gemini）による日本語翻訳・解説です。臨床判断・引用の際は必ず原文をご確認ください。
原文URL: https://doi.org/10.48683/1926.00129237

#2 【学術論文】 A Systemic Observational Framework Integrating Physiotherapy-Derived Functional, Physiological, And Behavioral Indicators In Predicting Pharmacological Response

著者: Dr. Kanchan Kholiya, Dr. Shalini dixit, Ram Sihag (PT), Dr. Samarpita Senapati (PT), Dr. Harish Sharma (PT), Dr. Gurpreet Kaur (PT)), Dr. Harshika Gupta (PT).	
	掲載誌: Cultura (2026-11-10)
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1. 論文・ニュース概要

None

2. 公開ステータス

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3. 英文アブストラクト（原文抄録）

Background: Inter-individual variability in pharmacological response remains a persistent challenge in clinical rehabilitation. While pharmacokinetic and molecular mechanisms are well established, they account for only a proportion of observed outcome variability. Functional performance, physiological regulation, and behavioral engagement-core domains of physiotherapy-are increasingly recognized as influential determinants of therapeutic response. Aim: To develop and describe a systemic observational framework integrating physiotherapy-derived functional, physiological, and behavioral indicators with pharmacological response, supported by descriptive statistical reasoning. Methodology: A conceptual observational methodology was adopted. Peer-reviewed literature was synthesized, and commonly reported trends, proportions, and associations were interpreted using descriptive and inferential statistical logic (e.g., variability, correlation strength, explained variance), without primary data collection. Results: Evidence across rehabilitation and pharmacological literature consistently indicates that functional and behavioral variables account for a substantial proportion of outcome variability unexplained by pharmacological parameters alone. Descriptive trends suggest moderate-to-strong associations between autonomic regulation, movement efficiency, behavioral engagement, and therapeutic stability. Conclusion: Incorporating physiotherapy-derived indicators into pharmacological interpretation enhances explanatory power and clinical relevance. A systemic, statistically informed observational framework supports personalized, interdisciplinary healthcare.

4. 原文確認のご案内

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原文URL: https://doi.org/10.5281/zenodo.18309329

#3 【学術論文】 Effects of Single-Session High-Definition Transcranial Alternating Current Stimulation (HD-tACS) Combined with Robot-Assisted Hand Exercises on Corticospinal Excitability in Chronic Stroke

著者: Meng H, Houston M, Rao AZ, Rho J	掲載誌: Am J Phys Med Rehabil (2026 Nov 1)
DOI: 10.1097/phm.0000000000003108	区分: 抄録公開

1. 論文・ニュース概要

None

2. 公開ステータス

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- 掲載誌: Am J Phys Med Rehabil
- 出版日: 2026 Nov 1

3. 英文アブストラクト（原文抄録）

OBJECTIVE: This preliminary study investigated whether a single session of high-definition transcranial alternating current stimulation (HD-tACS) could improve corticospinal tract (CST) excitability compared with sham stimulation in healthy individuals and individuals with chronic stroke.

DESIGN: Twelve healthy adults and 7 individuals with chronic stroke received active and sham HD-tACS over the primary motor cortex (M1) flexor digitorum superficialis (FDS) hotspot, ipsilesional in stroke subjects, for 30 minutes at 2mA and 21 Hz in a randomized crossover design. All subjects performed robot-assisted grasp-and-open hand exercises during active and sham stimulation.

RESULTS: HD-tACS significantly increased FDS MEP amplitudes in the healthy group (15.3%). In the stroke group, FDS MEP amplitudes were increased by ~22.3% after HD-tACS, but this change was not statistically significant. When MEP amplitudes were normalized as percentage change, HD-tACS significantly increased FDS MEP amplitudes in both groups compared with the sham condition. No significant extensor digitorum communis (EDC) modulation was observed in either group.

CONCLUSION: HD-tACS combined with robot-assisted hand exercises may enhance CST excitability compared with robot-assisted hand exercises alone after a single session in both healthy and stroke subjects. However, larger studies are needed to confirm these findings.

4. 原文確認のご案内

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原文URL: https://doi.org/10.1097/PHM.0000000000003108

#4 【学術論文】 Self-reported difficulty reading small print and in-hospital behavioral change in older adults with cardiovascular disease: Implications for post-discharge cardiovascular outcomes

著者: Ogura A, Izawa KP, Kanejima Y, Kitamura M	掲載誌: Geriatr Nurs (2026 Oct 16)
DOI: 10.1016/j.gerinurse.2026.104390	区分: 抄録公開

1. 論文・ニュース概要

None

2. 公開ステータス

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・出版日: 2026 Oct 16

3. 英文アブストラクト（原文抄録）

OBJECTIVES: Visual dysfunction may hinder patient education and self-care in older patients with cardiovascular disease (CVD); however, its association with behavioral change during hospitalization and subsequent cardiovascular prognosis remains unclear. This study investigated whether self-reported difficulty reading small print is associated with transtheoretical model (TTM) stage progression during hospitalization and whether such change predicts post-discharge outcomes.

METHODS: This prospective multicenter cohort study enrolled older patients who underwent cardiac rehabilitation during hospitalization. Visual dysfunction was assessed by self-reporting of difficulty reading small print. Behavioral change was evaluated using the TTM. The primary outcome was unplanned rehospitalization due to major adverse cardiovascular and cerebrovascular events (MACCE) within one year. Multivariable logistic regression identified factors associated with TTM improvement, and Cox proportional hazards models were applied to examine the association between TTM improvement and MACCE rehospitalization.

RESULTS: Among 351 included patients, 31.9% had difficulty reading small print. After multivariable adjustment, difficulty reading small print was independently associated with lower odds of TTM improvement (odds ratio, 0.31; p=0.006). Patients achieving TTM improvement had a significantly lower risk of MACCE rehospitalization after adjustment for age, sex, institution, heart failure, renal disease, and TTM stage at discharge (hazard ratio, 0.37; p = 0.042).

CONCLUSIONS: Self-reported difficulty reading small print was independently associated with reduced TTM stage improvement during hospitalization, and TTM stage improvement predicted a lower risk of MACCE rehospitalization. These findings highlight the importance of focusing on the behavioral change process during hospitalization and incorporating visual considerations into behavioral support strategies for secondary prevention in older CVD patients.

4. 原文確認のご案内

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原文URL: https://doi.org/10.1016/j.gerinurse.2026.104390

#5 【厚労省通知】 薬事審議会 医療機器・体外診断薬部会を開催します

著者: 情報なし	掲載誌: 掲載誌不明 (2026-10-16)
DOI: なし	区分: 抄録公開

1. 論文・ニュース概要

None

2. 公開ステータス

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- 掲載誌: 情報なし
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